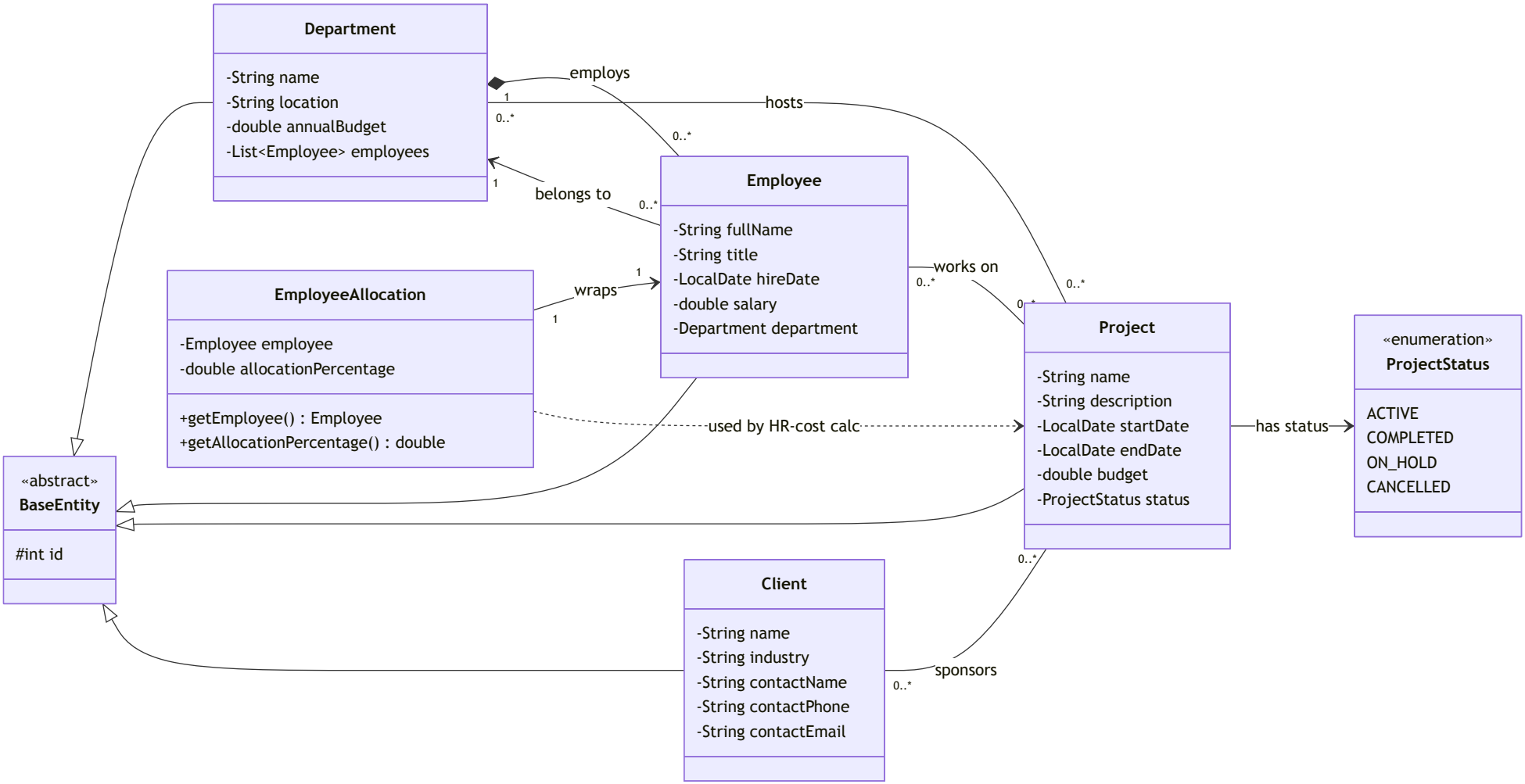
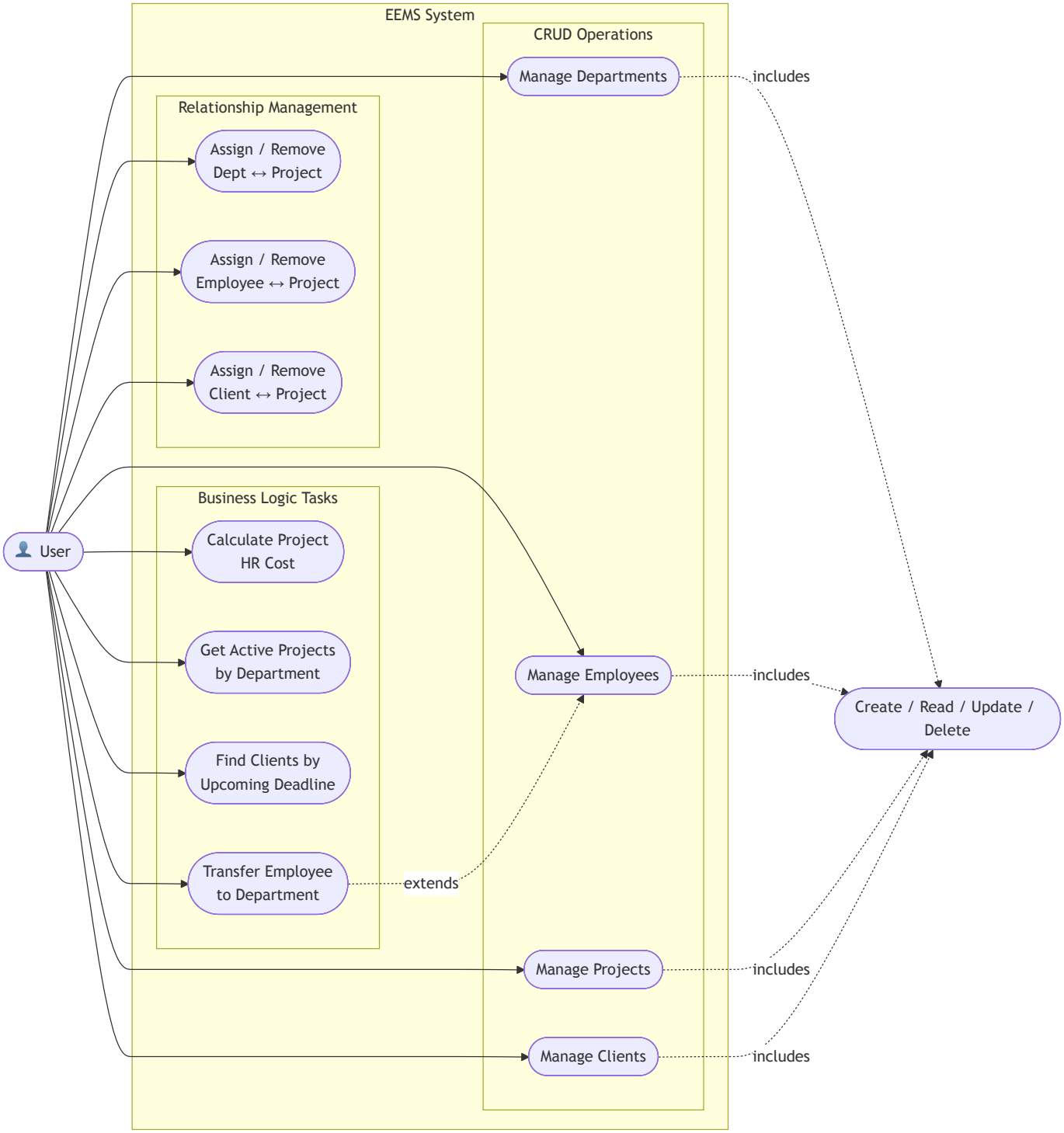
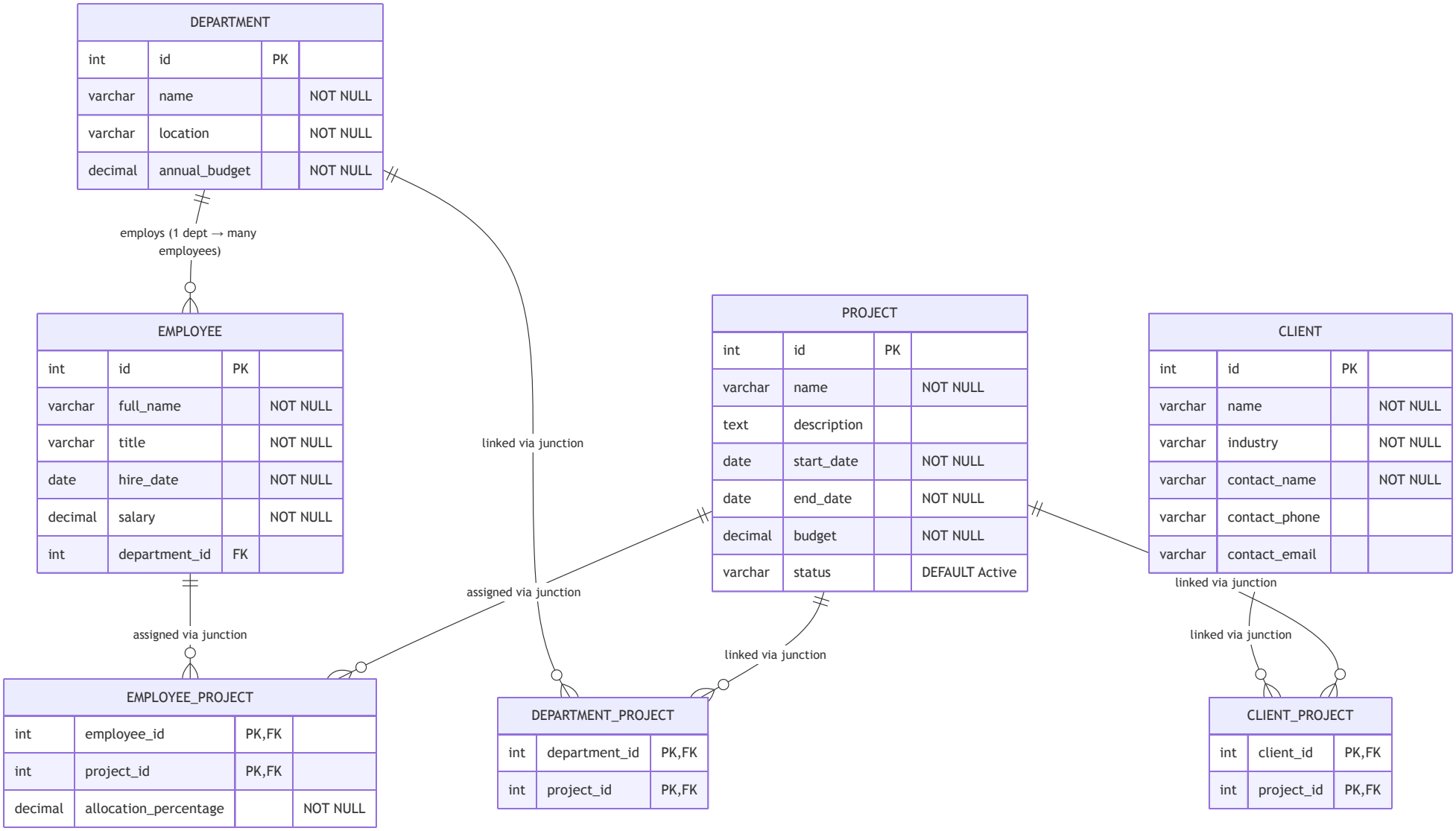
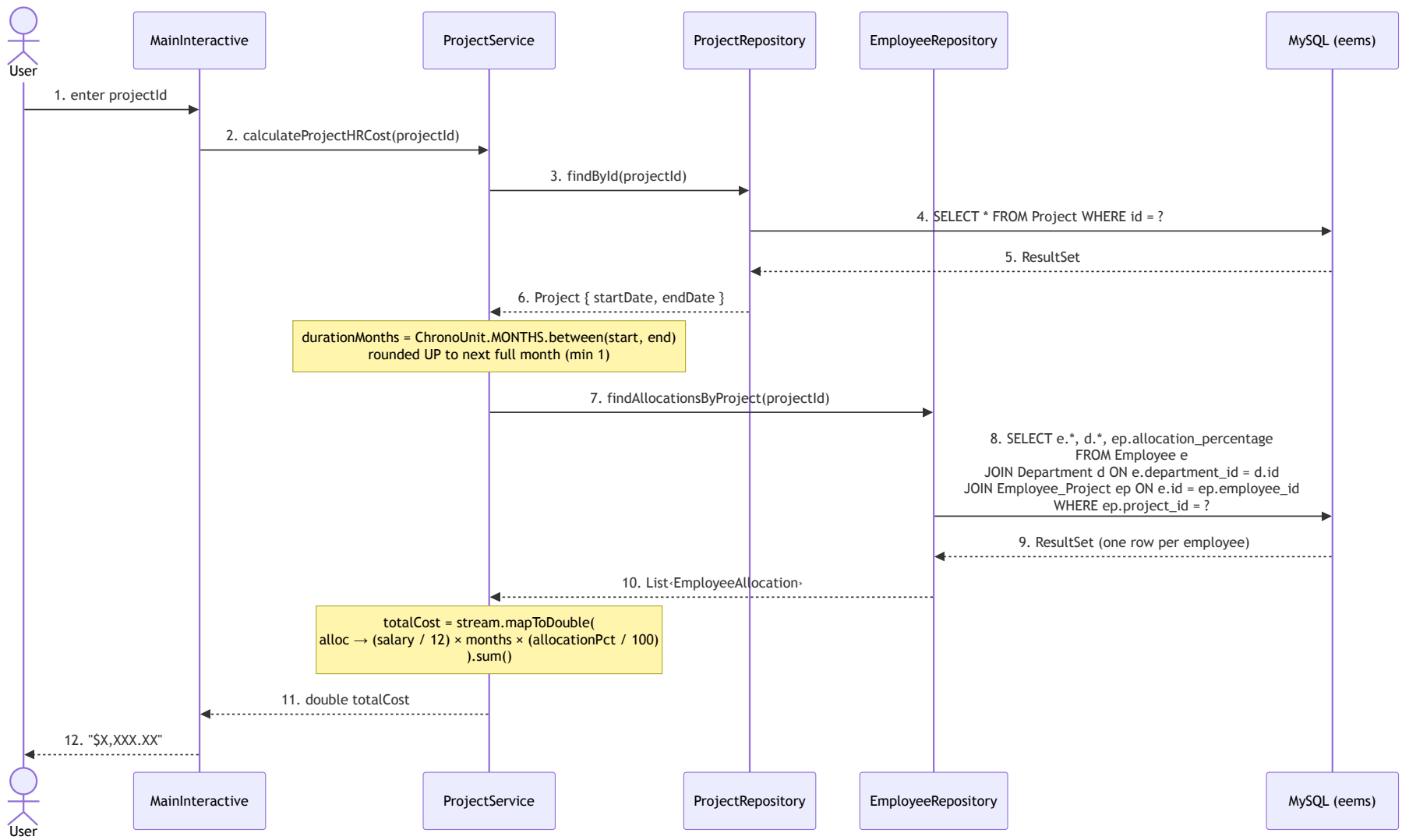

Department
-String name
-String location
-double annualBudget
-List<Employee> employees
employs
hosts
belongs to
1
0..*
Employee
-String fullName
-String title
-LocalDate hireDate
-double salary
-Department department
works on
EmployeeAllocation
-Employee employee
-double allocationPercentage
+getEmployee() : Employee
+getAllocationPercentage() : double
wraps
used by HR-cost calc
Project
-String name
-String description
-LocalDate startDate
-LocalDate endDate
-double budget
-ProjectStatus status
has status
«enumeration»
ProjectStatus
ACTIVE
COMPLETED
ON_HOLD
CANCELLED
«abstract»
BaseEntity
#int id
Client
-String name
-String industry
-String contactName
-String contactPhone
-String contactEmail
sponsors


EEMS System
CRUD Operations
Manage Departments
includes
Relationship Management
Assign / Remove Dept ↔ Project
Assign / Remove Employee ↔ Project
Assign / Remove Client ↔ Project
Business Logic Tasks
User
Calculate Project HR Cost
Get Active Projects by Department
Find Clients by Upcoming Deadline
Transfer Employee to Department
extends
Manage Employees
includes
Create / Read / Update / Delete
Manage Projects
includes
Manage Clients
includes


DEPARTMENT
int id PK
varchar name NOT NULL
varchar location NOT NULL
decimal annual_budget NOT NULL
employs (1 dept → many employees)
EMPLOYEE
int id PK
varchar full_name NOT NULL
varchar title NOT NULL
date hire_date NOT NULL
decimal salary NOT NULL
int department_id FK
assigned via junction
EMPLOYEE_PROJECT
int employee_id PK,FK
int project_id PK,FK
decimal allocation_percentage NOT NULL
linked via junction
assigned via junction
PROJECT
int id PK
varchar name NOT NULL
text description
date start_date NOT NULL
date end_date NOT NULL
decimal budget NOT NULL
varchar status DEFAULT Active
linked via junction
DEPARTMENT_PROJECT
int department_id PK,FK
int project_id PK,FK
CLIENT
int id PK
varchar name NOT NULL
varchar industry NOT NULL
varchar contact_name NOT NULL
varchar contact_phone
varchar contact_email
linked via junction
linked via junction
CLIENT_PROJECT
int client_id PK,FK
int project_id PK,FK


User
MainInteractive
ProjectService
ProjectRepository
EmployeeRepository
MySQL (eems)
1. enter projectId
2. calculateProjectHRCost(projectId)
3. findById(projectId)
4. SELECT * FROM Project WHERE id = ?
5. ResultSet
6. Project { startDate, endDate }
durationMonths = ChronoUnit.MONTHS.between(start, end)
rounded UP to next full month (min 1)
7. findAllocationsByProject(projectId)
8. SELECT e.*, d.*, ep.allocation_percentage
FROM Employee e
JOIN Department d ON e.department_id = d.id
JOIN Employee_Project ep ON e.id = ep.employee_id
WHERE ep.project_id = ?
9. ResultSet (one row per employee)
10. List‹EmployeeAllocation›
totalCost = stream.mapToDouble(
alloc → (salary / 12) × months × (allocationPct / 100)
).sum()
11. double totalCost
12. "$X,XXX.XX"
User
MainInteractive
ProjectService
ProjectRepository
EmployeeRepository
MySQL (eems)